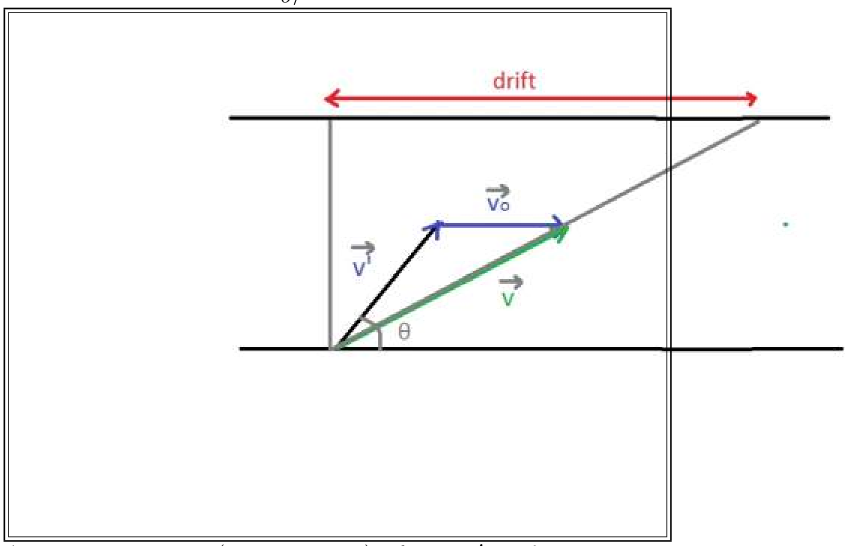
Question: A boat moves relative to water with a velocity which is n = 2.0 times less than the river flow velocity. At what angle to the stream direction must the boat move to minimize drifting?

Solution: Let the river flow velocity $= v_o$. and velocity of boat relative to water would be then $v' = v_o/n$.



Across component(along y axis) of v is $v' \sin\theta$

Drift component(along x axis) of v is $v' \cos\vartheta + v_o$

drift $= \dfrac{AB}{v'\sin\theta}\left(v'\cos\vartheta + v_o\right) = AB\left(\cot\theta + \dfrac{v_o}{v'}cosec\theta\right) = AB\left(\cot\theta + ncosec\theta\right)$

Now we have to minimize drift,

At minima, $\dfrac{d}{d\theta}drift = 0$

$\Rightarrow \dfrac{d}{d\theta}AB\left(\cot\theta + ncosec\theta\right) = 0$

$\Rightarrow AB(cosec^2\theta + ncosec\theta\cot\theta) = 0$

$\Rightarrow \cos\vartheta = -\dfrac{1}{n}$

The only solution in (0,π) is 120^o.